The Larger Picture: A Designerly Approach to Making the Invisible Domestic Workloads of Working Women Visible

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ABSTRACT

The pandemic in 2020 and the resultant global lockdown led to new dynamics in the homestead. Families faced an accelerated process of digitization of conventional processes like education and white collar jobs. Working women with children of school or kindergarten age were forced to redraw the boundaries between the personal and the professional as the entire family essentially functioned from home, which increased prevalent gender inequalities. Tasks like 'planning', care-giving, emotional care etc. go unaccounted for and remain invisible to other members of the household. We present a feminist design research project that explores how design research can contribute to greater social justice and gender equality by a fairer distribution of domestic labour. We conducted semi-structured interviews in India during the pandemic lockdown, to elicit both explicit as well as tacit information on daily workloads of working women with families. We found disparity in domestic duties between genders, with specific regard to the circumstances created by the 2020 pandemic conditions. Based on our findings and analysis, we prototyped a visualization tool that makes invisible domestic work visible. Our research contribution is a design research methodology to generate insights to enable creation of design interventions for social justice. This was achieved through use of established design research methods, that are conventionally used in face-to-face settings, which were then adapted to exceptional pandemic conditions that required social distancing where communication was remote, and usually virtual.

CCS CONCEPTS

• **Social and professional topics** → User characteristics; Gender; Women; • **Human-centered computing** → Interaction design; Interaction design process and methods; Scenario-based design.

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KEYWORDS

Invisible Work, Design Research Methods, Household Gender Parity

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1 INTRODUCTION

Families have had to face a sudden and abrupt process of digital adoption of conventional structures of work and education due to the 2020 pandemic. In the new family dynamic, the boundaries between the personal and the professional blurred with the entire family being homebound. Working women with young children took up additional domestic duties in the new normal. Traditional societies like India have had unequal gender care burdens [1], [2]. The 80's and the 90's saw increased participation of women in the workforce but their care burdens remained undiminished [3], [4]. The pandemic pushed the scales further askew with women taking on additional duties at home in the time that was earlier spent away in offices [5]–[8].

Domestic work is often invisible or not valued by virtue of being unpaid work. Thus, it is not included in the Gross Domestic Product (GDP), although it is the prerequisite of the productive labour sector [9]. Domestic work is visible or invisible in a society due to many cultural, social and conventional factors. The indicators that help make some of this work visible result from social negotiations, social portrayal and the linked presumptions [10]. Women at home have been essential cogs in the social network that is perpetuated by systemic control and relegation of domestic duties [11].

This study was conducted as part of a course where user-centered and participatory methods were deployed to discover and modify socio-technical inequalities exacerbated by the pandemic crisis. The aim was to identify and attempt to improve issues faced by vulnerable marginalized social groups through technological solutions, deploying design as a tool for social justice [12]. Feminist standpoint theory requires us to reflect on the situation of our knowledge as an embodied experience [12]. We decided to look at working middle class women with young children as a marginalized group, as all authors share a feminist view, acutely aware of unequal distribution of household work in the Indian society, where working women still shoulder an unequal burden of household work and care duties alongside their professional responsibilities. The specific research question we ask is how to overcome or reduce social inequality using design research methods and design interventions. Our research contribution is a design research methodology to generate insights to enable creation of design interventions for social justice.

We approached the design challenge of making invisible work visible through a Feminist Science, Technology & Society (FTTS) approach [13]–[17]. We conducted interviews and identified categories of invisible work. Our findings validated our assumptions about unequal distribution of domestic work. Based on the analysis of the semi-structured interviews and brainstorming, we proposed a visualization tool that helped us ideate and generate a possible design intervention that would aid in making invisible work, visible and could also help in scheduling.

2 BACKGROUND

2.1 Feminist Approaches to STS

Butler [18] defines gender as 'a performance or social achievement, constructed in interaction', a dynamic construction of identity that is shaped by technology. Feminist research has conceptualized technology as culture [8, 9], with a focus on cultural representation and consumption.

Feminist studies of technology trace their roots to Marxist debates on labour and production that posited that class conflict moulds technology [19]. Feminist sociologies argued that production of work due to technological change was driven by gender divisions as much as class ones.

Mainstream metrics like the GDP neglected to recognize the unpaid domestic labour done by women in homes, contributing to gender oppression. The sexual division of labour reinforced the divide between women and control of technologies, in both the professional and domestic spheres, further restricting their agency [20], [21]. Gender has been theorized to be the make or break point of a future utopia...or dystopia [22], [23]. Silverstone and Hirsch [24] further posited that the character of domesticity itself is transforming and the new social construction of 'family' will contribute significantly to this break point. The pandemic precipitated one such unforeseen and unprecedented break point. This created a unique simulation of what happens to the social construct under stress.

Star [2, 13] proposed a feminist theory in knowledge representation, pointing out that women have developed a skill-set to multitask that is essential to any venture but is ignored as formal work. Women have thus become simultaneous insiders and outsiders in knowledge work. Biesecker and von Winterfeld [26], [27] proposed a new social contract between genders for any solution to be sustainable. In the case of professional women, this social contract affects both professional and domestic duties.

2.2 Invisible Work and its Categorization

Allen [16:4] defines visible work as 'formal work that is authorized and documented', relegating care work and domestic chores to the invisible category. Invisible not just to outsiders, but very often, to friends and family as well [10], [29]. Star and Strauss [10] stressed that work was rendered visible or invisible through both tangible and intangible indicators, along with context. In fact, the irony may well be that, 'the better the work is done, the less visible it is to those who benefit from it' [30:58]. Representations then get stereotyped, and the more distant the observer is, the more simple and insignificant the work seems. So to recognize invisible work, we must be able to formalize it and classify it in some way.

Bowker and Star [31] defined classification as 'a spatial, temporal, or spatio-temporal segmentation of the world'. They posit "to classify is human"; that classifications may often be ad hoc because categorization is a natural impulse of humans and do not necessarily need to be standardized or documented.

They admit that the ideal scheme does not exist due to the trade-offs of reality that need to be negotiated. This is where marginalized work and workers slip through the cracks. Large data systems have been notorious for quashing the voice and value of gender with layers of infrastructure [32]. Studies have shown that intersectional variations like generation, race, socio-economic status etc compound the gender gap in creation and delivery of products and services [17, 21]. A universal classification is difficult to crystallize but defining a specific user group like working middle class women with young children can help narrow the focus, as tried to do in our study.

2.3 Making the Invisible Visible through ICTs

Simone de Beauvoir [34] said, 'Representation of the world, like the world itself, is the work of men; they describe it from their own point of view, which they confuse with the absolute truth.' Accurate visualization then becomes crucial to an equitable representation. Representation is meant to give a voice to the invisible built through engagement for wider portrayal [30].

In 'Invisible Women', Caroline Criado Perez [9] highlights the complete absence of women in the cultural and social chronicling of humanity. The unfair, even if unintended, bias has led to a society built by men, for men. The burgeoning digital technologies have afforded novel methods of research and design solutions with quick turnaround times. The 'Virtual Society?' by the British Economic and Social Research Council described how social behaviour, social organization and even social interaction have been shaped by new electronic technologies [35]. More recently, digital tools have shaped the methods of ethnographic data collection [36] [37]. The unprecedented 2020 Corona Pandemic and the subsequent lockdown created conditions for women at home that exacerbated the existing issues of gender inequality, both at home [38] and at work [39], unequal economic opportunities [40] and their well-being [41]. In the Indian context too, issues of structural inequality and societal skewed gender norms were heightened [5]–[8]. In this context, we conducted this design research within the constraints dictated by the pandemic and its restrictions.

3 METHODS

This research was done during a course titled "Designing for (post-)pandemic societies". We were to apply user-centered and participatory design methods for feminist ends to discover and modify socio-technical inequalities increased by the current pandemic crisis, and/or to improve the situation of marginalized/vulnerable social groups by technological solutions. Hence the methods that we chose for the study were pre-determined by the course structure.

We conducted semi-structured ethnographic interviews amongst Indian women with young children to identify the unequal distribution of domestic chores as a result of the Corona pandemic that forced most members of the household to function from home. Due to the unique circumstances created by a global lockdown, our study was conducted virtually. Household data regarding work distribution, types of work done and pandemic related changes were compiled from existing literature and semi-structured interviews. We analyzed this data to generate scenarios to help identify and define the target users' problems. These scenarios, and further brainstorming, helped in generating a categorized list of household work. The list was then validated using an online survey. Based on analysis of results of the survey, scenario and interviews, we conceptualized a possible tool that might help raise awareness about the prevalent unequal distribution of work in Indian middle class households, by making the invisible work, visible.

3.1 Understanding the problem space

We studied existing literature focusing on unequal work distribution in a household, issues faced by working women, and pandemic related changes. The learnings from this study were used to visualize the various levels of stakeholders concerning working women with young children (Figure 1). We also generated mind maps to understand aspects of a family (Figure 2). We planned semi-structured interviews with intended users to understand the domain, by using a participatory study approach.

3.2 Creating scenarios to understand the target population

Using the mind maps and literature analysis, we prepared interview probes that we categorized into three major segments, 1) Personal space, time and relationships 2) Pandemic related changes and 3) Economics, social class, privilege. We then conducted six semi-structured interviews with working women having children younger than 10 years. Informed consent was taken for the same and all data was de-identified before use. The observations and responses were grouped into three segments and analyzed. The analysis of these interviews, and the literature study, helped us identify two broad categories of the target population. We then generated an actual scenario for each of these categories. We used scenario generation as a method as these example stories helped us understand the context, stakeholders and critical factors affecting the problem. This helped us identify avenues for design intervention and provided a structure for further brainstorming and ideation.

3.3 Identification and categorization of household work

To identify the 'invisible work' done by women in the household, we explored the various chores in the household and generated an exhaustive list of tasks and categorized them broadly in terms of daily, seasonal and emergency work. This list was then verified using an online survey of 10 women from the user group we were looking at. Validating the categories helped us avoid an ad hoc creation of classification [31].The objective of the survey was

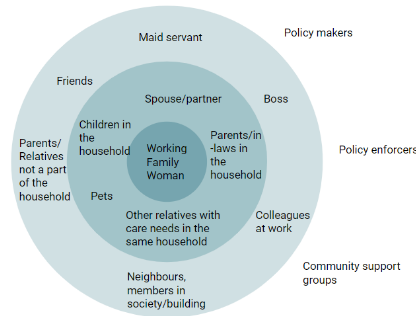

Figure 1: Stakeholder map

to 1) verify the identified tasks, 2) understand the distribution of these tasks within household members, and, 3) check if there is any discomfort or reluctance to identify the distribution of work and whether it leads to any triggers for further conversation. We surveyed ten women from different parts of India. We examined the responses and generated a data visualization detailing the overall distribution of work (Figure 3). This visualization was then used as a prompt to ideate a possible design intervention using a critical design approach.

3.4 Generating a concept for probable design intervention

We brainstormed for various possible uses of ICT to create conversation triggers and generate awareness about the possibility of unequal distribution of work in the household. After multiple brainstorming and ideation sessions, we identified four major features that were required in our proposed intervention. These features included, 1) A method of listing and identifying work in the household in order to schedule and distribute it, 2) Data visualization of work distribution in a particular household and a general visualization with anonymized data, 3) A platform for sharing the visualizations and solutions these women have found to the problem of scheduling and work distributions, both locally and globally, and 4) A reward system to incentivize and gamify the distribution and handling of chores to encourage other household members to participate.

We then created some parts of information architecture (Figure 4-6) and quick wireframes (Figure 7), to incorporate and visualize some aspects of the proposed design intervention.

4 FINDINGS

4.1 Stakeholder map and mind-maps to understand the wider context

As an outcome of initial literature study and brainstorming to understand the target audience, we generated a stakeholder map (Figure 1) to depict a network of people in the target individual's immediate psychosocial field and included household members as well as others who have a significant impact. Figure 2 below depicts the mind map that was then generated to depict the aspects of what it means to be a family.

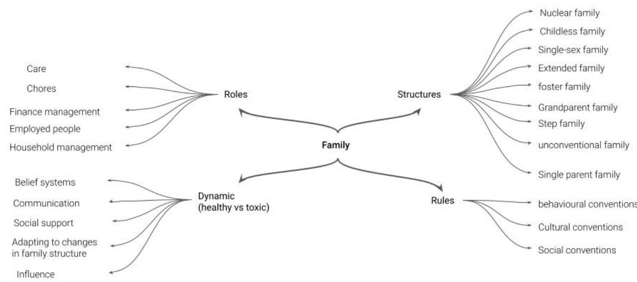

Figure 2: Mind map depicting multiple aspects of a family

Table 1: Current Scenarios

Actual Scenario #1: Traditional work distribution	Actual Scenario #2: Comparable work distribution
Spouse (usually husband) earns more, believes in traditional distribution of work and responsibilities	Spouse participates equally in decision making and work at the household
Spouse (usually husband) takes all major house-related and financial decisions	Balanced distribution of roles and responsibilities
No support from family for working outside, expectations, no appreciation or acknowledgement of unpaid labour	Support from family for paid labour
Woman have to partially sacrifice her work for household responsibilities, loss of financial independence	Woman still plays a bigger role out of choice in childcare and child's education
Skewed workload because of belief in traditional gender roles	Manages to find leisure time for herself (but might spend it in planning chores or work)
Added responsibilities because of Corona - educating the child, taking care of everyone's needs and demands, lack of house-help	Flexible sharing of household work
No time for herself, however has accepted her roles and fate due to social conditioning	Worried about family's health, screen time
Worried about family's health, screen time	

4.2 Findings from the semi- structured interviews and generation of scenarios

The findings form semi-structured interviews were classified in three major categories:

1) Personal space, time and relationships

- Family is placed before career in all cases
- Husbands did not feature much in the conversations of those who were shouldering most of the domestic duties
- Children not involved in domestic chores; their education has high priority

2) Pandemic related changes

- Appreciate the additional time with children and increased dialogue/leisure time together
- They were worried about child's increased screen time

3) Economics, social class, privilege.

- Woman have chosen the kind of paid work which also allows them to balance household duties especially care duties
- Education levels of both partners were comparable
- Quiet acceptance that domestic chores are their tasks

These findings as well as study of literature allowed us to identify two broad categories of target population. These categories were then described in form of the following two actual scenarios and a possible future scenario.

4.3 Validation & visualisation of household work

We created a list of household work and classified it under categories of regular, seasonal and emergency work, depending on occurrence and exigency. We then validated the list through a user survey. The outcome of 10 surveys was visualized as a bar chart to visualize distribution of work where the overall approximate contribution of wife, spouse and others could be gauged (Figure 3).

4.4 Proposing intervention for making invisible work visible

We then ideated on possible intervention that could aid as a trigger/prompt for generating awareness and also as an interesting way of scheduling and distribution of household work by adding elements of gamification like a reward system. We propose the

Figure 3: Visualization to show overall distribution of work in the household

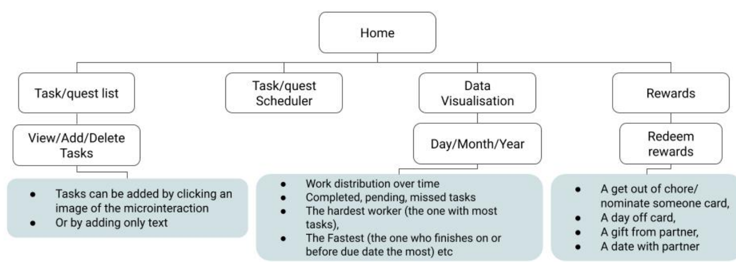

Figure 4: The home page and first level of menu for the primary stakeholders

following information architecture as a base for a website or an application (Figure 4-6) and have created a mock-up wireframe of what such an app could look like (Figure 7) with a focus on visualization and reward system.

5 DISCUSSION

> "*...any social system (including an organization or a country or a family) is the way it is because the people in that system (at least those individuals and factions with the most leverage) want it that way.*" [42]

When we started this research project, we had two major assumptions:

1. There is an unequal distribution of work in the household
2. This issue of unequal distribution is primarily a problem of planning and scheduling work.

After we verified the first assumption using interviews, we developed a planning and scheduling solution. However, after analyzing the interviews and existing literature, we realized that most of the women in these situations are unaware that the system is dysfunctional or are resigned to this system with a view that this has been and will be the norm. The interviews as well as the survey helped us understand the extent to which household work is unequally distributed. It also helped us generate scenarios that aided the ideation process.

Then, our design intervention's overarching purpose shifted from solving an identified technical problem of scheduling to identifying and tackling an adaptive challenge of creating awareness about the unequal distribution of invisible household work, and we decided to use a critical design approach. The information we collected from the probes enabled us to categorize the different kinds of invisible work, towards the goal of making it visible. Awareness of these invisible tasks is the first step in recognizing the unequal domestic burdens shared by partners in the house, which was exacerbated by the new normal of global lockdowns and novel work/study from home processes.

Instead of trying to solve the scheduling and planning problem, we felt that it would be essential to try and tackle the underlying

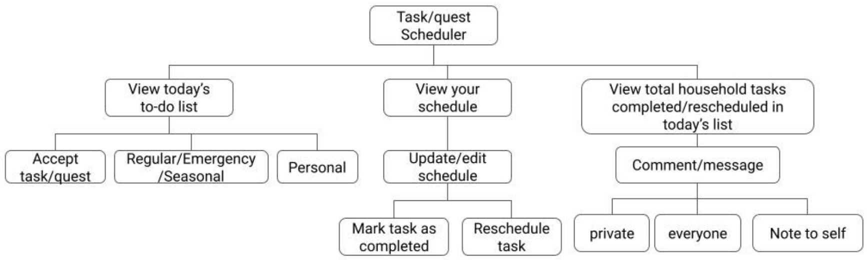

Figure 5: The options and actions possible under section of scheduler

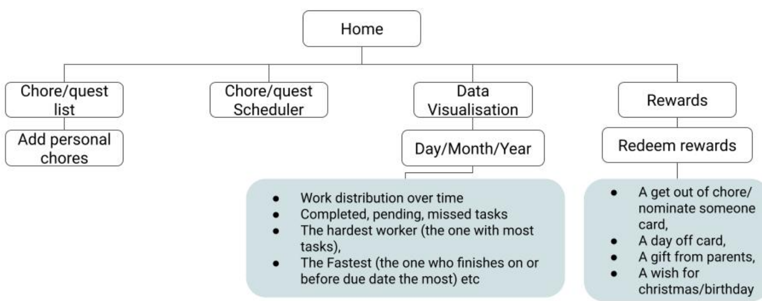

Figure 6: The options and actions available for secondary stakeholders in the household

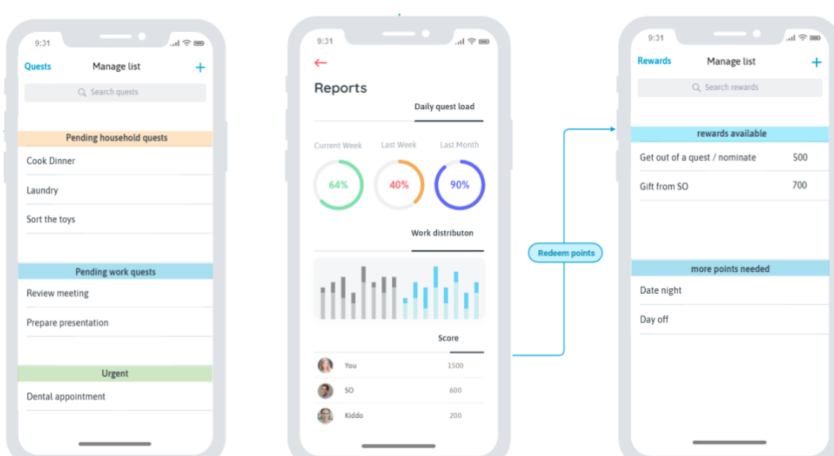

Figure 7: Mock up wireframe of a possible app

structural issue of social conditioning and patriarchal mindset. The objective was to generate an intervention that would trigger or act as a prompt for meaningful reflection and expose the hidden socio-cultural norms that have led to the current status quo and work distribution in an Indian household. We have decided to use visualization of work distribution to provoke and thus enable reaction, reflection and hopefully subsequent change in the society.

Though the context of this research was Indian, we feel that inequity in distribution of unpaid household labour is a global trend due to existing social structures. We feel that the proposed intervention can be easily scaled to include invisible household labour across countries and cultures with some modifications.

6 CONCLUSION

Our study showed that the circumstances created by the Corona pandemic exacerbated the structural problems of skewed gender roles in the household. For working women, the invisible load of additional chores and child care, sharing time and 'space' with their 'work from home' existence, led to a double burden. Women are aware of this invisible work to a very limited extent and largely perceive it as a part of their duty and responsibility. The challenge then becomes to identify and design interventions that increase women's critical awareness of this inequality. Our interviews showed that it is usually a case of un-willful omission and bringing the lapses to light can help bring balance to the household. However since the interviews were conducted before avenues of design intervention were considered, we feel the next set of primary research should also focus on classifying the participants on the basis of specifications of class, family structure, age and income. Our design solution proposes a detailed categorization and further visualization of these invisible tasks in a household, to initiate discussion among the family members that could break the cycle of continued gender inequalities. Our study admittedly was restricted to a certain demographic so future work could involve survey deployment and prototype validation across a larger and more heterogeneous sample. The gamified version of this visualization tool also holds promise in creating awareness, starting a dialogue, and making the invisible domestic workloads of working women visible.

The results were only related to an individual, concentrated private circle. Finding individual solutions for structural problems is generally difficult to implement. One possibility for increasing the scope of observation would be to share the processed survey-outcomes. This would make it possible to establish a sense of togetherness and because of the community level, a self-awareness of the unequal distribution.

REFERENCES

[1] Usha Rani Rout, Sue Lewis, and Carolyn Kagan. 1999. Work and Family Roles: Indian Career Women in India and the West. Indian Journal of Gender Studies 6, 1: 91–105. https://doi.org/10.1177/097152159900600106

[2] Radhika Garg and Subhasree Sengupta. 2019. When you can do it, why can't I?: Racial and Socioeconomic Differences in Family Technology Use and Non-Use. Proceedings of the ACM on Human-Computer Interaction 3, CSCW: 1–22. https://doi.org/10.1145

[3] Suman Verma and Reed Larson. 2001. Indian women's experience of household labour: Oppression or personal fulfillment. *Indian Journal of Social Work* 62 (2001): 46-66. TISS.EDU

[4] G.N. Ramu. 1987. Indian husbands: Their role perceptions and performance in single-and dual-earner families. *Journal of Marriage and the Family*, pp.903-915. https://www.jstor.org/stable/pdf/351983.pdf?casa_token=dYX5SKBMFWgAAAAA:VUgCMKxbd38C1R2wUc2UukP1XfcFezrVE5VK1ymSzh-A892FoqMosSJwHV7abtLEXHQH2dHRwZ5NTuL_l5_Z6qu7jKf9KNnE4YeW4n3JfChYsuXOskIpeQ.

[5] Debadrita Chakraborty. 2020. The "living dead" within "death-worlds": Gender crisis and covid-19 in India. Gender, Work & Organization. https://doi.org/10.1111/gwao.12585

[6] Priyanshi Chauhan. 2020. Gendering COVID-19: Impact of the Pandemic on Women's Burden of Unpaid Work in India. Gender Issues. https://doi.org/10.1007/s12147-020-09269-w

[7] Nitya Rao and Jayashree Balasubramanian. 2020. An unequal pandemic. NatureIndia. Published online 23 September 2020 doi:10.1038/nindia.2020.145

[8] Ashwini Deshpande. 2020. The COVID-19 Pandemic and Gendered Division of Paid and Unpaid Work: Evidence from India OCTOBER 2020 IZA – Institute of Labor Economics IZA DP No. 13815 ISSN: 2365-9793

[9] Caroline Criado-Perez. 2019. Invisible women : data bias in a world designed for men. New York: Abrams Press xv 411 pages

[10] Susan Leigh Star and Anselm Strauss. 1999. Layers of Silence, Arenas of Voice: The Ecology of Visible and Invisible Work. Computer Supported Cooperative Work (CSCW) 8, 1–2: 9–30. https://doi.org/10.1023/a:1008651105359

[11] Susan Leigh Star. 1990. Power, Technology and the Phenomenology of Conventions: On being Allergic to Onions. The Sociological Review 38, 1_suppl: 26–56. https://doi.org/10.1111/j.1467-954x.1990.tb03347.x

[12] Sasha Costanza-Chock. 2020. *Design justice: Community-led practices to build the worlds we need.* The MIT Press. Cambridge, MA OAPEN.ORG

[13] Cynthia Cockburn & Susan Ormrod. 1993. Gender and technology in the making. London: Sage Publications, Inc. Google Books

[14] Donald MacKenzie& Judy Wajcman. 1999. The Social Shaping of Technoloy. (2nd Edition). Buckingham: Open University Press. https://eprints.lse.ac.uk/28638/

[15] Judy Wajcman., *Technofeminism*. Cambridge: Polity Press ISBN: 0-7456-3043-X (cart.).

[16] Corinna Bath, 2013. Searching For Methodology. In Gender in Science and Technology. transcript-Verlag, 57–78. https://doi.org/10.14361/transcript.9783839424346.57

[17] Els Rommes, Corinna Bath, and Susanne Maass. 2012. Methods for Intervention. Science, Technology, & Human Values 37, 6: 653–662. https://doi.org/10.1177/0162243912450343

[18] Judith Butler, *Gender trouble, feminist theory, and psychoanalytic discourse.* Feminism/postmodernism 327 (1990): x. Google Books

[19] Judy Wajcman. 2000. Reflections on Gender and Technology Studies: Social Studies of Science 30, 3: 447–464. https://doi.org/10.1177/030631200030003005

[20] Judy Wajcman. 2002. Addressing Technological Change: The Challenge to Social Theory.*Current Sociology* 50, no. 3 (May 2002): 347–63. https://doi.org/10.1177/0011392102050003004.

[21] Judy Wajcman. 2010. Feminist theories of technology. Cambridge Journal of Economics 34, 1: 143–152. https://doi.org/10.1093/cje/ben057

[22] Howard Rheingold. 1994. The Virtual Community. (New York: Harper, 1994), 1994. Google Books

[23] Nicholas Negroponte. 1995. *Being Digital*. Sydney: Hodder & Stoughton, 1995. http://web.stanford.edu/class/sts175/NewFiles/Negroponte.%20Being%20Digital.pdf

[24] Roger Silverstone and Eric Hirsch. 1992. Introduction, in Silverstone and Hirsch (eds), in *Consuming Technologies: Media and Information in Domestic Spaces*, (London: Routledge, 1992), 1992, pp. 1–11. Google Books

[25] Susan Leigh Star. 1991. The sociology of the invisible: The primacy of work in the writings of Anselm Strauss., in *Social organization and social process: Essays in honor of Anselm Strauss*, 1991, p. pp.265-283. Google Books

[26] Adelheid Biesecker and Sabine Hofmeister. 2010. Focus: (Re)productivity. Ecological Economics 69, 8: 1703–1711. https://doi.org/10.1016/j.ecolecon.2010.03.025

[27] Adelheid Biesecker and Uta von Winterfeld. 2017. Notion of multiple crisis and feminist perspectives on social contract. Gender, Work & Organization 25, 3: 279–293. https://doi.org/10.1111/gwao.12206

[28] Davina Allen. 2014. The Invisible Work of Nurses. Routledge. https://doi.org/10.4324/9781315857794

[29] Doris Lydahl. 2017. Visible persons, invisible work? Exploring articulation work in the implementation of person-centred care on a hospital ward, *Sociol. Forsk.*, vol. 54, no. 3, pp. 163–179, 2017. https://www.jstor.org/stable/26632245?seq=1#metadata_info_tab_contents

[30] Lucy Suchman. 2016. Making work visible, *New Prod. Users Chang. Innov. Collect. Involv. Strateg.*, pp. 125–135, 2016. Google Books

[31] Geoffrey C. Bowker, Susan Leigh Star. 2000. Sorting Things Out: Classification and Its Consequences, MIT Press. Google Books

[32] Geoffrey C. Bowker, Stefan Timmermans, and Susan Leigh Star. 1996. Infrastructure and Organizational Transformation: Classifying Nurses' Work. In Information Technology and Changes in Organizational Work. Springer US, 344–370. https://doi.org/10.1007/978-0-387-34872-8_21

[33] Sandra Buchmüller, Gesche Joost, Nina Bessing, and Stephanie Stein. 2011. Bridging the gender and generation gap by ICT applying a participatory design process. Personal and Ubiquitous Computing 15, 7: 743–758. https://doi.org/10.1007/s00779-011-0388-y
[34] Simone de Beauvoir, *The Second Sex*. 1949.
[35] Sonia Liff. 1990. Stunted Growth or Slow Development? The Coverage of Gender Issues within ESRC-Funded Research on InformationTechnology, 1990.
[36] Sarah Pink, Heather Horst, John Postill, Larissa Hjorth, Tania Lewis and Jo Tacchi. 2016. Digital Ethnography: Principles and Practice. London, UK : SAGE Publications Ltd, 2016. 202 p.
[37] Shaowen Bardzell. 2010. Feminist HCI. In Proceedings of the 28th international conference on Human factors in computing systems - CHI '10. https://doi.org/10.1145/1753326.1753521
[38] Tali Kristal and Meir Yaish. 2020. Does the coronavirus pandemic level the gender inequality curve? (It doesn't). Research in Social Stratification and Mobility 68: 100520. https://doi.org/10.1016/j.rssm.2020.100520
[39] Caitlyn Collins, Liana Christin Landivar, Leah Ruppanner, and William J. Scarborough. 2020. COVID-19 and the gender gap in work hours. Gender, Work & Organization 28, S1: 101–112. https://doi.org/10.1111/gwao.12506
[40] Michèle Tertilt, Matthias Doepke, Titan Alon and Jane Olmstead-Rumsey. 2020. This time it's different: the role of women's employment in a pandemic recession. 2020. Centre for Economic Policy Research, 33 Great Sutton Street, London EC1V 0DX, UK https://repec.cepr.org/repec/cpr/ceprdp/DP15149.pdf
[41] Ben Etheridge and Lisa Spantig. 2020. The Gender Gap in Mental Well-Being During the Covid-19 Outbreak: Evidence from the UK. 2020. http://hdl.handle.net/10419/227789
[42] Ronald Heifetz, Alexander Grashow, and Marty Linsky. 2009. Leadership in a (permanent) crisis. *Harv. Bus. Rev.*, vol. 87, no. 7/, pp. 62–69, 2009. https://hbr.org/2009/07/leadership-in-a-permanent-crisis